

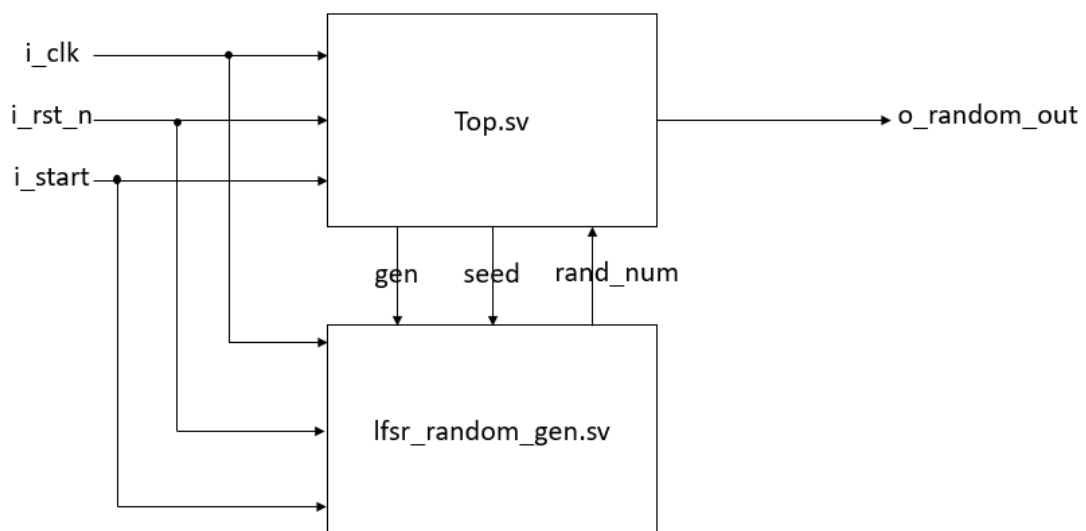
數位電路實驗 Lab1 team9

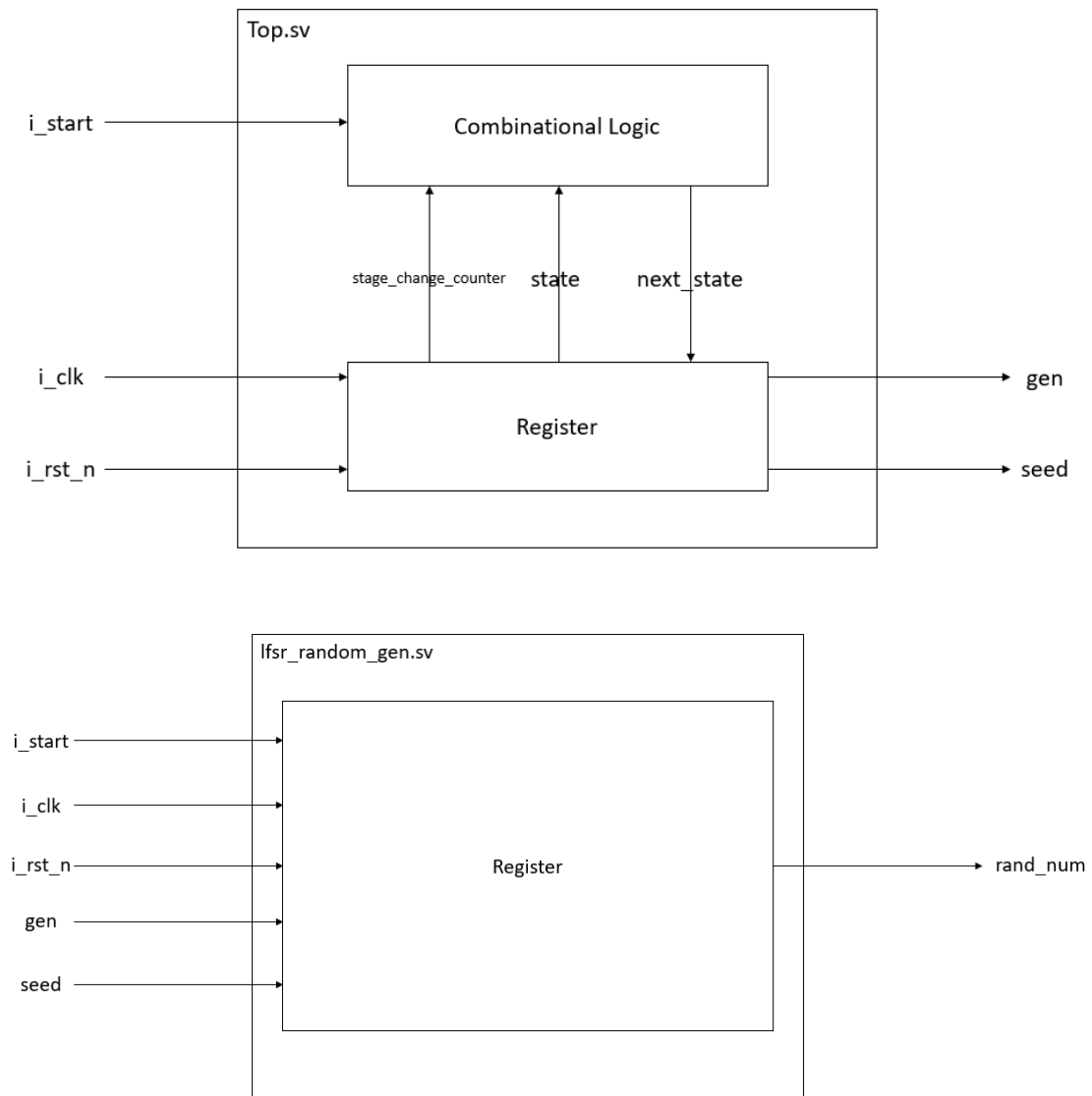
B12901005 郭力銘 B12901185 謝承恩 B12901191 彭子綸

1. File Structure

```
team09_lab1
|- team09_lab1_report.pdf
|- src
    |- DE2_115
        |- DE2_115.qsf
        |- DE2_115.sdc
        |- DE2_115.sv
        |- Debounce.sv
        |- SevenHexDecoder.sv
    |- Top.sv
    |- lfsr_random_gen.sv
```

2. System Architecture



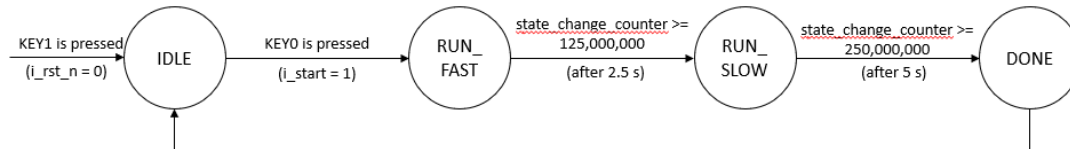


Variable Explanation

Variable	Explanation
i_clk	Clock.
i_rst_n	KEY1, asynchronous reset.
i_start	KEY0, press to start.
gen	When gen = 1, random num is generated.
seed	Provide the random num a seed to randomized.
state	Remember the state now.
next_state	Determine the next state.
counter(in reg, not shown in the pic)	Check whether to change the output.
stage_change_counter	Check whether to change state.

rand_num	The num generated by lfsr.
o_random_out	Output.

3. Hardware Scheduling



State transition explanation

State	Explanation
IDLE	Wait for KEY0 to be pressed
RUN_FAST	Output random num with the frequency of 5Hz
RUN_SLOW	Output random num with the frequency of 2Hz
DONE	Hold the final output

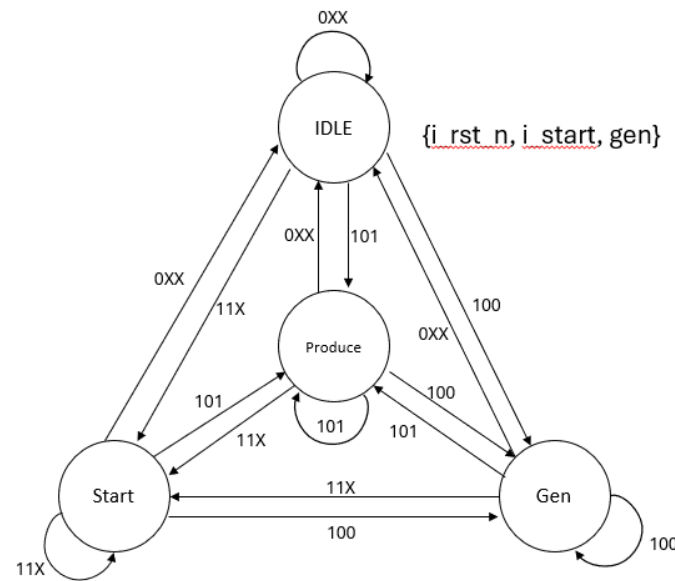
When in IDLE stage, we wait for KEY0. As KEY0 to be pressed, the state changes to RUN_FAST to produce a 5Hz random num signal, which is calculated by “counter” to determine whether to output. After 2.5 seconds, calculated by “state_change_counter”, the state changes to RUN_SLOW, to produce a 2Hz random num signal. Finally, after 2.5 seconds more, which is also calculated by “state_change_counter”, the state changes to DONE, and hence hold the output.

Some tricks

(1) We design the output frequency to be 5Hz or 2Hz. Each state holds for 2.5 seconds. Hence, in the design, every time when counter = $50,000,000 * 0.2 = 10,000,000$ in 5Hz state, and counter = $50,000,000 * 0.5 = 25,000,000$ in 2Hz state, we change the output. Also, when state_change_counter = $50,000,000 * 2.5 = 125,000,000$, we change the state from RUN_FAST to RUN_SLOW. And when state_change_counter = $50,000,000 * 5 = 250,000,000$, we change the state from RUN_SLOW to DONE.

(2) When KEY1 being pressed, which means to reset, we design the seed to be reset to 0, too. Then, as the clock goes on, we make the seed to plus 1 after every clock cycle. Hence, the seed will be different determined by when you press KEY0.

Operational logic in lfsr_random_gen:



State	Explanation
IDLE	Initialize all register in lfsr_random_gen
Start	Start to produce the random number
Gen	Store the previous random number
Produce	Produce the next random number by LFSR and other operations

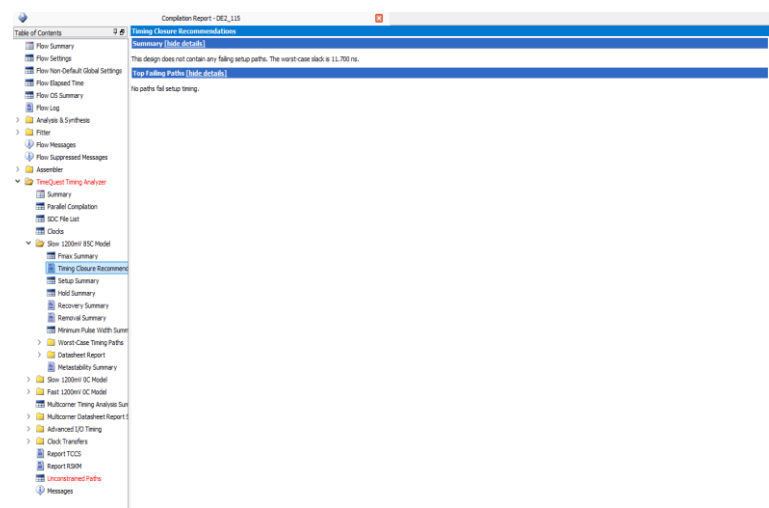
4. Fitter Summary

Fitter Summary	
Fitter Status	Successful - Thu Sep 18 16:01:08 2025
Quartus II 64-Bit Version	15.0.0 Build 145 04/22/2015 SJ Full Version
Revision Name	DE2_115
Top-level Entity Name	DE2_115
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	268 / 114,480 (< 1 %)
Total combinational functions	239 / 114,480 (< 1 %)
Dedicated logic registers	165 / 114,480 (< 1 %)
Total registers	165
Total pins	518 / 529 (98 %)
Total virtual pins	0
Total memory bits	0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements	2 / 532 (< 1 %)
Total PLLs	0 / 4 (0 %)

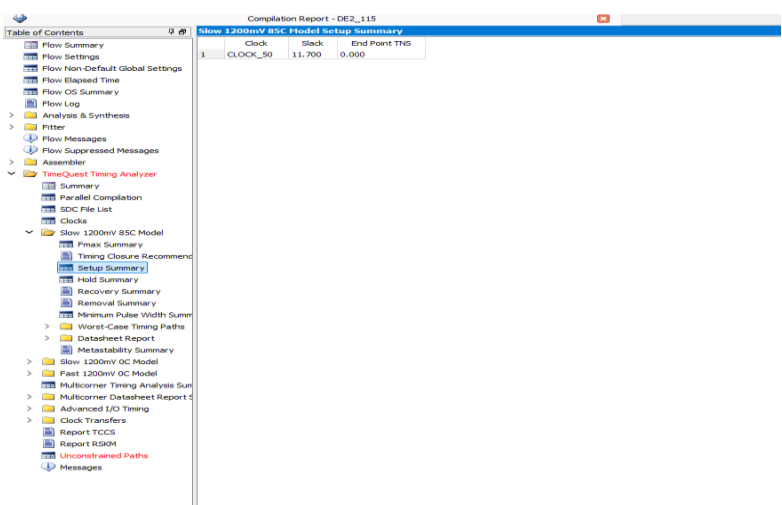
5. Timing Analyzer

(1) Slow 1200mV 85C Model

Timing closure recommendations



Setup summary



Minimum Pulse Width Summary

Compilation Report - DE2_115				
Slow 1200mV 85C Model Minimum Pulse Width Summary				
	Clock	Slack	End Point TNS	
1	CLOCK_50	9.413	0.000	
2	CLOCK2_50	16.000	0.000	
3	CLOCK3_50	16.000	0.000	

(2) Worst-case Timing Paths

Setup CLOCK_50

Compilation Report - DE2_115									
Slow 1200mV 85C Model Setup: 'CLOCK_50'									
	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay	
1	11.700	Top:tap0[counter][3]	Top:tap0[counter][0]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
2	11.700	Top:tap0[counter][3]	Top:tap0[counter][1]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
3	11.700	Top:tap0[counter][3]	Top:tap0[counter][2]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
4	11.700	Top:tap0[counter][3]	Top:tap0[counter][3]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
5	11.700	Top:tap0[counter][3]	Top:tap0[counter][4]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
6	11.700	Top:tap0[counter][3]	Top:tap0[counter][5]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
7	11.700	Top:tap0[counter][3]	Top:tap0[counter][6]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
8	11.700	Top:tap0[counter][3]	Top:tap0[counter][7]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
9	11.700	Top:tap0[counter][3]	Top:tap0[counter][8]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
10	11.700	Top:tap0[counter][3]	Top:tap0[counter][9]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
11	11.700	Top:tap0[counter][3]	Top:tap0[counter][10]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
12	11.700	Top:tap0[counter][3]	Top:tap0[counter][11]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
13	11.700	Top:tap0[counter][3]	Top:tap0[counter][12]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
14	11.700	Top:tap0[counter][3]	Top:tap0[counter][13]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
15	11.700	Top:tap0[counter][3]	Top:tap0[counter][14]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
16	11.700	Top:tap0[counter][3]	Top:tap0[counter][15]	CLOCK_50	CLOCK_50	20,000	-0.079	8.219	
17	11.791	Top:tap0[counter][3]	Top:tap0[counter][25]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
18	11.791	Top:tap0[counter][3]	Top:tap0[counter][24]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
19	11.791	Top:tap0[counter][3]	Top:tap0[counter][16]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
20	11.791	Top:tap0[counter][3]	Top:tap0[counter][17]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
21	11.791	Top:tap0[counter][3]	Top:tap0[counter][18]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
22	11.791	Top:tap0[counter][3]	Top:tap0[counter][19]	Top:tap0[counter][17]_50	CLOCK_50	20,000	-0.063	8.144	
23	11.791	Top:tap0[counter][3]	Top:tap0[counter][20]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
24	11.791	Top:tap0[counter][3]	Top:tap0[counter][21]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
25	11.791	Top:tap0[counter][3]	Top:tap0[counter][22]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
26	11.791	Top:tap0[counter][3]	Top:tap0[counter][23]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
27	11.791	Top:tap0[counter][3]	Top:tap0[counter][29]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
28	11.791	Top:tap0[counter][3]	Top:tap0[counter][26]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
29	11.791	Top:tap0[counter][3]	Top:tap0[counter][27]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
30	11.791	Top:tap0[counter][3]	Top:tap0[counter][28]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
31	11.791	Top:tap0[counter][3]	Top:tap0[counter][31]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
32	11.791	Top:tap0[counter][3]	Top:tap0[counter][30]	CLOCK_50	CLOCK_50	20,000	-0.063	8.144	
33	11.866	Top:tap0[counter][2]	Top:tap0[counter][0]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
34	11.866	Top:tap0[counter][2]	Top:tap0[counter][1]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
35	11.866	Top:tap0[counter][2]	Top:tap0[counter][2]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
36	11.866	Top:tap0[counter][2]	Top:tap0[counter][3]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
37	11.866	Top:tap0[counter][2]	Top:tap0[counter][4]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
38	11.866	Top:tap0[counter][2]	Top:tap0[counter][5]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
39	11.866	Top:tap0[counter][2]	Top:tap0[counter][6]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
40	11.866	Top:tap0[counter][2]	Top:tap0[counter][7]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
41	11.866	Top:tap0[counter][2]	Top:tap0[counter][8]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
42	11.866	Top:tap0[counter][2]	Top:tap0[counter][9]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
43	11.866	Top:tap0[counter][2]	Top:tap0[counter][10]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
44	11.866	Top:tap0[counter][2]	Top:tap0[counter][11]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
45	11.866	Top:tap0[counter][2]	Top:tap0[counter][12]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
46	11.866	Top:tap0[counter][2]	Top:tap0[counter][13]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
47	11.866	Top:tap0[counter][2]	Top:tap0[counter][14]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
48	11.866	Top:tap0[counter][2]	Top:tap0[counter][15]	CLOCK_50	CLOCK_50	20,000	-0.079	8.053	
49	11.970	Top:tap0[counter][2]	Top:tap0[counter][25]	CLOCK_50	CLOCK_50	20,000	-0.063	7.965	
50	11.970	Top:tap0[counter][2]	Top:tap0[counter][24]	CLOCK_50	CLOCK_50	20,000	-0.063	7.965	
51	11.970	Top:tap0[counter][2]	Top:tap0[counter][16]	CLOCK_50	CLOCK_50	20,000	-0.063	7.965	

Compilation Report - DE2_115								
Slow 1200mV 85C Model Minimum Pulse Width: 'CLOCK_50'								
Table of Contents	Back	Actual Width	Required Width	Type	Clock	Clock Edge	Target	
Flow Summary	50	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c1]
Flow Settings	51	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c13]
Flow Non-Default Global Settings	52	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c14]
Flow Elapsed Time	53	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c15]
Flow OS Summary	54	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c16]
Flow Log	55	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c17]
Analysis & Synthesis	56	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c18]
Fitter	57	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c19]
Flow Messages	58	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c20]
Flow Suppressed Messages	59	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c21]
Assembler	60	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c22]
TimeQuest Timing Analyzer	61	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c23]
Summary	62	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c24]
Parallel Compilation	63	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c25]
SDC File List	64	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c26]
Clocks	65	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c27]
Slow 1200mV 85C Model	66	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c28]
Fmax Summary	67	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c29]
Timing Closure Recommendations	68	9.675	9.863	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c30]
Setup Summary	69	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c31]
Hold Summary	70	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c32]
Recovery Summary	71	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c33]
Removal Summary	72	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c34]
Minimum Pulse Width Summary	73	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c35]
Worst-Case Timing Paths	74	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c36]
Setup: 'CLOCK_50'	75	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c37]
Hold: 'CLOCK_50'	76	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c38]
Minimum Pulse Width: 'CLOCK_50'	77	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c39]
Minimum Pulse Width: 'CLOCK_50'	78	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c40]
Minimum Pulse Width: 'CLOCK_50'	79	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c41]
Minimum Pulse Width: 'CLOCK_50'	80	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c42]
Minimum Pulse Width: 'CLOCK_50'	81	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c43]
Minimum Pulse Width: 'CLOCK_50'	82	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c44]
Minimum Pulse Width: 'CLOCK_50'	83	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c45]
Minimum Pulse Width: 'CLOCK_50'	84	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c46]
Minimum Pulse Width: 'CLOCK_50'	85	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c47]
Minimum Pulse Width: 'CLOCK_50'	86	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c48]
Minimum Pulse Width: 'CLOCK_50'	87	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c49]
Minimum Pulse Width: 'CLOCK_50'	88	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c50]
Minimum Pulse Width: 'CLOCK_50'	89	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c51]
Minimum Pulse Width: 'CLOCK_50'	90	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c52]
Minimum Pulse Width: 'CLOCK_50'	91	9.677	9.865	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c53]
Minimum Pulse Width: 'CLOCK_50'	92	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c54]
Minimum Pulse Width: 'CLOCK_50'	93	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c55]
Minimum Pulse Width: 'CLOCK_50'	94	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c56]
Minimum Pulse Width: 'CLOCK_50'	95	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c57]
Minimum Pulse Width: 'CLOCK_50'	96	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c58]
Minimum Pulse Width: 'CLOCK_50'	97	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c59]
Minimum Pulse Width: 'CLOCK_50'	98	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c60]
Minimum Pulse Width: 'CLOCK_50'	99	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c61]
Minimum Pulse Width: 'CLOCK_50'	100	9.678	9.866	0.188	Low Pulse Width	CLOCK_50	Rise	Tap:tap0[iffr_random_genmu_andiffr_c62]

(3) Fast 1200mV OC Model

Multicorner Timing Analysis Summary

Compilation Report - DE2_115

Multicorner Timing Analysis Summary

	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	Worst-case Slack	11.700	0.181	N/A	N/A	9.239
1	CLOCK2_50	N/A	N/A	N/A	N/A	16.000
2	CLOCK3_50	N/A	N/A	N/A	N/A	16.000
3	CLOCK_50	11.700	0.181	N/A	N/A	9.239
2	Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	CLOCK2_50	N/A	N/A	N/A	N/A	0.000
2	CLOCK3_50	N/A	N/A	N/A	N/A	0.000
3	CLOCK_50	0.000	0.000	N/A	N/A	0.000

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 - Hold: 'CLOCK_50'
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 - Minimum Pulse Width: 'CLOCK3_50'
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 - Metastability Summary
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 - Fast 1200mV OC Model
 - Multicorner Timing Analysis Summary
 - Multicorner Datasheet Report Summary
 - Advanced I/O Timing
 - Clock Transfers
 - Report TCOS
 - Report RSKM
 - Unconstrained Paths
 - Messages

Setup Summary

Compilation Report - DE2_115

Fast 1200mV 0C Model Setup Summary			
	Clock	Slack	End Point TNS
1	CLOCK_50	16.021	0.000

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- Flow Summary
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Minimum Pulse Width Summary

Compilation Report - DE2_115

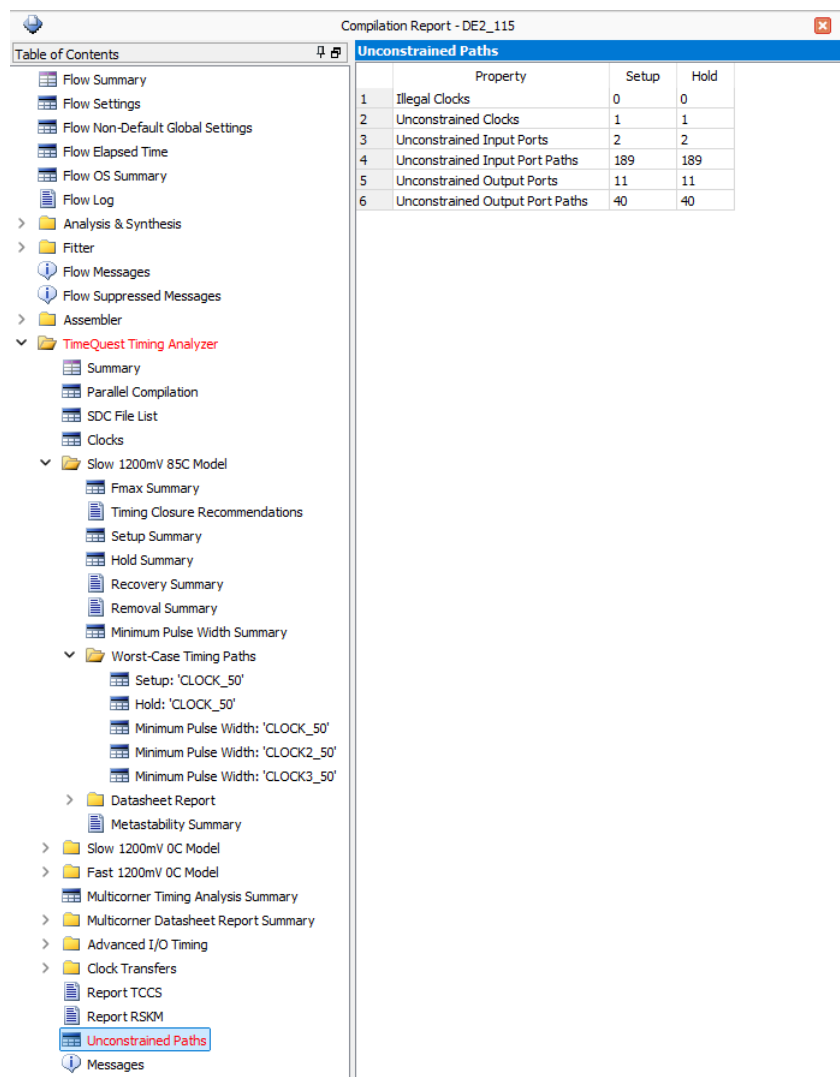
Fast 1200mV 0C Model Minimum Pulse Width Summary			
	Clock	Slack	End Point TNS
1	CLOCK_50	9.239	0.000
2	CLOCK2_50	16.000	0.000
3	CLOCK3_50	16.000	0.000

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(4) Clock Transfers

Unconstrained Paths



Unconstrained Paths			
	Property	Setup	Hold
1	Illegal Clocks	0	0
2	Unconstrained Clocks	1	1
3	Unconstrained Input Ports	2	2
4	Unconstrained Input Port Paths	189	189
5	Unconstrained Output Ports	11	11
6	Unconstrained Output Port Paths	40	40

6. 遇到的問題與心得

本次為我們第一次接觸 Verilog，我們大都沒有基礎，是從暑假開始慢慢自學起來的，途中當然遇到了許多問題，畢竟 Verilog 與其他的程式語言不同，著重描述了硬體的接線設計，與 C++ 或 python 可以使用變數隨意賦值的方式不同，在初期非常容易出錯，debug 也花了很久時間。這次 Lab 中遇到最大的問題應該是忘記初始化的部分，在設計 random output 的時候，由於針對每個變數都忘記初始化，跑 nWave 的時候一直出現 X，後來才發現問題所在。而其實環境架設也花了我們蠻大一段時間的，包括從 wsl 連接到學校 ip，使用 vcs 與 nWave 操作，還需要將本地資料上傳至 ip 位置，整套流程下來真的花了我們不少時間。最後看到成品可以照常跑出真的蠻感動的，途中也如前述的學到很多。